

Association of GLP-1 receptor agonists versus bariatric surgery with outcomes in obese patients with hypertrophic cardiomyopathy: a multicenter propensity-matched analysis

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Aims

Obesity is linked with worse outcomes in hypertrophic cardiomyopathy (HCM). While both bariatric surgery (BS) and GLP-1 receptor agonists (GLP-1RA) promote weight loss and may improve cardiometabolic risk, their comparative outcomes in HCM remain unclear.

Methods and results

Adults with HCM and body mass index (BMI) ≥ 30 kg/m were divided into two cohorts: (i) GLP-1 RA users who never had BS, and (ii) BS patients who never received GLP-1 RA. Propensity score matching (1:1) was performed to balance demographics, comorbidities, and medications. Outcomes over 24 months include all-cause mortality, acute heart failure (HF) hospitalization, arrhythmias, change in BMI, and cardiovascular symptoms. Hazard ratios with 95% confidence intervals were estimated using Cox models. Of 5002 patients, 955 were included in each matched cohort after matching (mean age 54 ± 13 years, 70% female, BMI 40.0 kg/m). GLP-1 RA was associated with a significantly lower observed rate of all-cause mortality compared with BS (2.0% vs. 4.4%; HR 0.46; 95% CI 0.27–0.80; $P = 0.004$) and HF hospitalization (8.7% vs. 14.0%; HR 0.60; 95% CI 0.46–0.79; $P < 0.001$). Similarly, GLP-1 RA use was associated with a lower incidence of composite cardiovascular outcomes (10.7% vs. 17.2%; HR 0.60; 95% CI 0.47–0.77; $P < 0.001$). Rates of arrhythmia and cardiovascular symptoms were comparable between the two groups. BMI reduction was greater after bariatric surgery (-4.0 vs. -2.6 kg/m; $P < 0.001$).

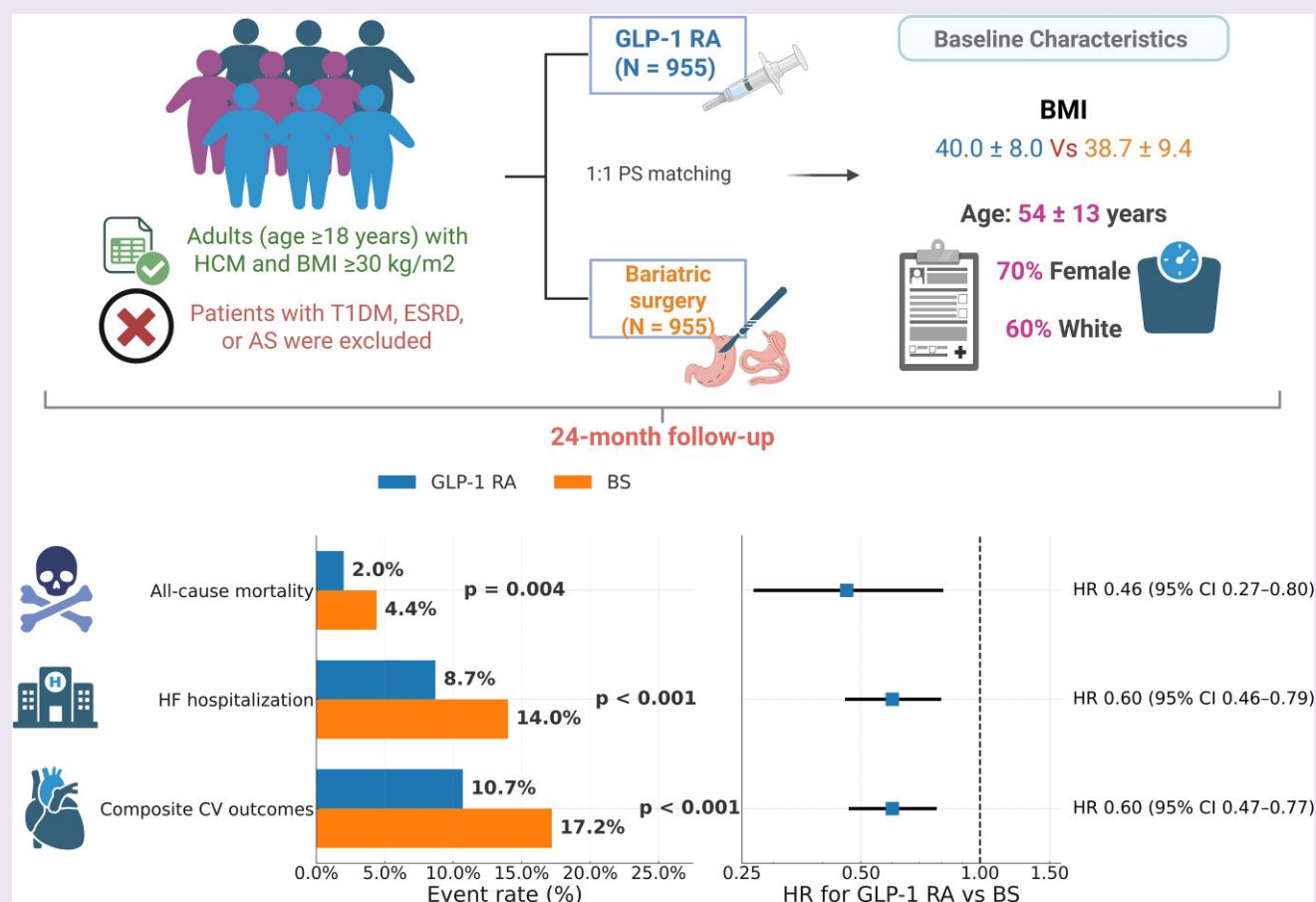
Conclusion

In obese patients with HCM, GLP-1 RA therapy was associated with lower all-cause mortality and fewer HF events compared with BS. Future studies are warranted to confirm our findings.

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Graphical abstract



Comparative outcomes of GLP-1 receptor agonists vs. bariatric surgery in patients with hypertrophic cardiomyopathy and obesity. AS = aortic stenosis; BMI = body mass index; BS = bariatric surgery; CI = confidence interval; CV = cardiovascular; ESRD = end-stage renal disease; GLP-1 RA = glucagon-like peptide-1 receptor agonist; HCM = hypertrophic cardiomyopathy; HF = heart failure; HR = hazard ratio; PS = propensity score; T1DM = type 1 diabetes mellitus.

Keywords

Hypertrophic cardiomyopathy • Heart failure • GLP-1 receptor agonist • Bariatric surgery • Obesity

Key Learning Points

What is already known

- Obesity is highly prevalent among patients with hypertrophic cardiomyopathy (HCM) and is associated with worse symptoms, greater left ventricular hypertrophy, and a higher risk of heart failure and adverse cardiovascular outcomes.
- While GLP-1 RA (receptor agonist) and bariatric surgery have each shown cardiovascular benefits in obese populations, their outcomes in HCM have not been evaluated.

What this study adds

- This multicentre, propensity-matched analysis is the first direct comparison between GLP-1 RA therapy and bariatric surgery in obese patients with HCM.
- GLP-1 RA use was associated with significantly lower all-cause mortality and fewer heart failure hospitalizations compared with bariatric surgery.
- However, weight loss achieved through GLP-1 RA and BS occurs via different pathways. GLP-1 RA was associated with more favourable outcomes despite less weight loss.

Introduction

Hypertrophic cardiomyopathy (HCM) is the most common genetic cardiomyopathy. HCM caused by sarcomeric protein mutations, characterized by asymmetric left ventricular (LV) hypertrophy not explained by other conditions, and is typically associated with preserved or increased ejection fraction.¹ Patients with HCM are frequently asymptomatic early in the disease, but with disease progression, they can develop dyspnoea, palpitations, angina, and/or syncope.² Comorbid conditions, such as obesity, can exacerbate symptoms and worsen prognosis.³ Obesity is common in patients with HCM, with rates over 40%.⁴

Obese patients with HCM are twice as likely to be symptomatic, more likely to progress to advanced stages of heart failure (HF), and have an increased risk of major adverse cardiovascular and cerebral events.⁵ These adverse outcomes occur due to obesity-related structural cardiac changes, including an increase in LV mass and LV cavity size, which worsen diastolic dysfunction.⁶ This underscores the critical importance of addressing obesity in patients with HCM. Current guidelines strongly recommend weight loss as part of the treatment strategy for individuals with HCM.² Among the available interventions, pharmacological treatment with glucagon-like peptide-1 receptor agonists (GLP-1 RA) and surgical intervention with bariatric surgeries (BS) have demonstrated promising outcomes.

GLP-1 RA not only aids in weight loss by delaying gastric emptying and enhancing satiety but also provides cardiometabolic benefit, including lowering blood pressure and improving diastolic function.⁷⁻⁹ In obese patients with HCM, studies have reported associations between GLP-1 RA use and lower all-cause mortality and HF hospitalizations, without increasing the risk of arrhythmias or drug-related adverse events.⁹ Similarly, BS is not only the most effective intervention for sustained weight loss, but it has been associated with favourable changes in cardiac function.¹⁰⁻¹² However, no direct head-to-head comparison between GLP-1 RA and BS in HCM patients have been conducted. Therefore, this study aimed to conduct a head-to-head comparison of the effects of GLP-1 RA and BS in obese patients with HCM, providing evidence to guide decision-making in this high-risk population.

Methods

Data source

This retrospective cohort study utilized the TriNetX Research Network database, which provides access to electronic health records (EHRs) from approximately 150 million individuals primarily in the U.S. This network provides data in an aggregate, deidentified manner to comply with the deidentification standard defined in Section §164.514(a) of the Health Insurance Portability and Accountability Act Privacy Rule. This study used anonymous patient data; hence, approval by the institutional review board was waived. The network was queried to identify the study cohorts by using the International Classification of Diseases-10th Revision (ICD-10) Clinical Modification, the ICD-10 Procedure Coding System, Current Procedural Terminology, and RxNorm.

Study population

This study included adults (age ≥ 18 years) who had a diagnosis of HCM (ICD-10 I42.1 or I42.2) and a body mass index (BMI) ≥ 30 kg/m² before September 2024. Patients with type 1 diabetes, end-stage renal disease, or aortic stenosis were excluded. The study participants were divided into two cohorts: (i) patients who received GLP-1 RA (exenatide, liraglutide, lixisenatide, albiglutide,

dulaglutide, semaglutide, or beinaglutide) within 12 months of HCM encounter, and (ii) those who underwent BS with no crossover permitted between groups. The index event was defined as 20 days after the bariatric surgery date to minimize the influence of peri-operative hospitalization and early postoperative complications. For the GLP-1 receptor agonist cohort, time zero was similarly defined as 20 days after the prescription date to account for potential delays between prescription and treatment initiation ([Figure 1](#)) ([Supplementary material online, Table S1](#)).

In addition, only patients with a documented diagnosis of HCM within the previous 12 months before the index event were included in each cohort, ensuring that all participants entered the study from a comparable, clinically relevant starting point and reducing the risk of immortal time bias as well as variability in disease severity. The study adhered to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines for reporting observational studies.

Clinical outcomes

The primary outcome were all-cause mortality, and composite outcomes were defined as HF exacerbation, all-cause mortality, ventricular arrhythmia, or cardiac arrest. While the secondary outcomes included HF exacerbation, all-cause hospitalization, new-onset atrial fibrillation/flutter, ventricular arrhythmia, cardiac arrest, cardiovascular symptoms (chest pain, abnormal breathing, palpitations, and/or lower extremity oedema), as well as change in BMI. Mortality data were derived from multiple sources, including health-care organization death records (HER-based), billable codes from closed claims, the Social Security Administration Master Death File, private obituaries, and private claims. HF exacerbation was defined as either a requirement for intravenous diuretic agents, an acute HF hospitalization, or a pulmonary oedema diagnosis. In addition, we evaluated safety outcomes, including gastrointestinal upset, gallbladder disease, and pancreatitis ([Supplementary material online, Table S2](#)).

Statistical analysis

Continuous variables were expressed as mean $\pm$ SD or median (IQR), and categorical variables were presented as number and percentage where appropriate. Baseline characteristics between the two cohorts were compared by using independent t-tests and chi-square tests for continuous and categorical variables, respectively. Propensity score matching (PSM) was performed using a 1:1 greedy nearest neighbour algorithm with a caliper of 0.1 pooled standardized mean difference (SMD). Covariates with an SMD < 0.1 were considered adequately matched. This approach mitigates baseline differences between the two study cohorts.

The following baseline characteristics were included in the PSM: demographics (age at index, race, sex, ethnicity, and BMI), baseline comorbidities [diabetes mellitus (DM), chronic kidney disease (CKD), essential hypertension, atrial fibrillation and flutter, history of ischaemic heart disease, history of cerebrovascular disease, dyslipidaemia, and tobacco smoking], medications (beta blockers, calcium channel blocker, amiodarone, insulin, oral hypoglycaemic agents, and diuretics), diagnostic values [low density lipoprotein (LDL), haemoglobin A1c, creatinine, and LV ejection fraction (LVEF)].

Cox regression hazard models were used to estimate hazard ratios (HR) with 95% confidence intervals (CI), respectively. Time-to-event analyses were conducted using Kaplan-Meier (KM) curves with log-rank tests to demonstrate the event probability difference between the two cohort. As part of our sensitivity analyses, we assessed the primary outcomes starting 3 months after the index event and calculated E-values for each endpoint. The significance threshold was established at $P < 0.05$. All analyses were conducted using complete-case data, as the TriNetX platform does not support imputation of missing laboratory or clinical variables. Statistical analyses were performed within the TriNetX Analytics Platform, and using R version 4.0.2 (R Foundation for Statistical Computing). Data analysis was conducted on 1 September 2024.

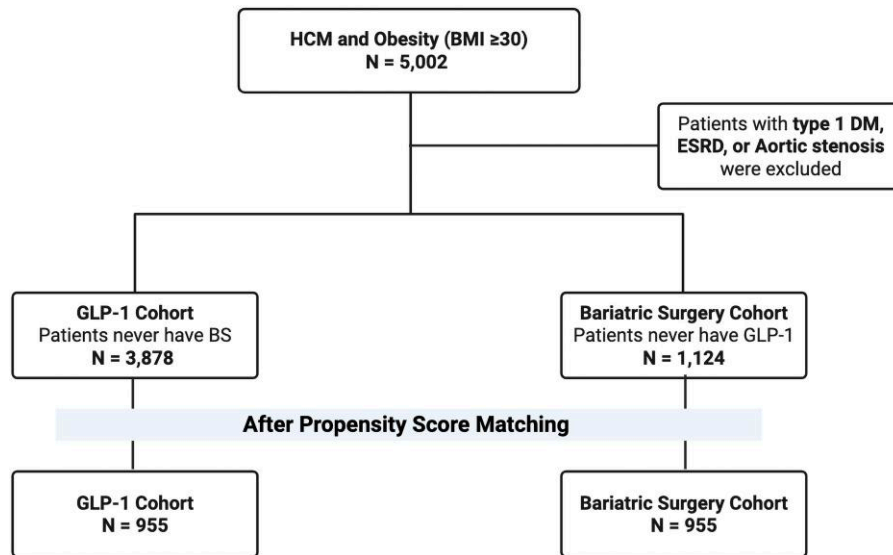


Figure 1 Flow chart to demonstrate the sample size, study population, and the inclusion criteria. HCM = hypertrophic cardiomyopathy; DM = diabetes mellitus; ESRD = end stage renal disease.

Results

Study population and baseline characteristics

A total of 5002 patients with HCM were identified, including 3878 in the GLP-1RA cohort and 1124 in the BS cohort. Patients in the GLP-1RA group were treated for a mean duration of 651 ± 147 days. Prior to matching, patients in the BS cohort were slightly younger (54.2 ± 12.8 vs. 57.6 ± 12.5 years, $P < 0.001$) and had a higher proportion of females (73.8% vs. 55.5%, $P < 0.001$). Race distribution was comparable between the groups, although non-Hispanic ethnicity was higher in the GLP-1 RA cohort (80.1% vs. 75.8%, $P = 0.002$). The GLP-1 RA cohort had higher prevalence of baseline comorbidities, including DM (66.6% vs. 25.9%, $P < 0.001$), CKD (19.9% vs. 10.5%, $P < 0.001$), hypertension (82.3% vs. 69.2%, $P < 0.001$), ischaemic heart disease (37.9% vs. 25.5%, $P < 0.001$), history of cerebral infarction (6.8% vs. 4.7%, $P = 0.010$), dyslipidaemia (75.6% vs. 49.3%, $P < 0.001$), and tobacco use (17.7% vs. 13.7%, $P = 0.002$) as compared to BS cohort.

Similarly, medication use was greater in the GLP-1 RA cohort, including beta-blockers (69.7% vs. 50.3%, $P < 0.001$), calcium channel blockers (48.7% vs. 30.2%, $P < 0.001$), diuretics (60.8% vs. 44.8%, $P < 0.001$), insulin (43.1% vs. 13.3%, $P < 0.001$), and oral hypoglycaemic agents (60.4% vs. 12.9%, $P < 0.001$). BMI was clinically similar (39.0 ± 9.3 vs. 38.1 ± 7.7 kg/m², $P = 0.004$), although more patients classified as ≥ 40 kg/m² were in the GLP-1 RA cohort (44.3% vs. 35.9%, $P < 0.001$) in the BS cohort. Bariatric patients had higher LDL (97.4 ± 37.1 vs. 92.8 ± 39.8 mg/dL, $P = 0.006$), whereas GLP-1 RA patients had higher haemoglobin A1c ($7.6 \pm 2.1\%$ vs. $5.9 \pm 1.2\%$, $P < 0.001$). LVEF was similar between the groups ([Table 1](#)).

After 1:1 PSM ([Supplementary material online, Figure S1](#)), 955 patients were included in each cohort. Baseline demographics and comorbidities were well balanced between the two cohorts with SMD < 0.1 across most characteristics. The mean age was 54 ± 13 years, with 70% female, 60% white, and 77% non-Hispanic in both cohorts. Comorbidities, including DM

(28%), CKD (11%), hypertension (71%), HF (28%), atrial fibrillation/flutter (20%), ischaemic heart disease (28%), dyslipidaemia (55%), and tobacco use (13%), were similar across cohorts. Medication use rates were comparable, including beta-blockers (55%), calcium channel blockers (33%), insulin (14%), oral hypoglycaemics (15%), diuretics (48%), and amiodarone (5%). BMI was (40.0 ± 8.0 in GLP-1 RA cohort vs. 38.7 ± 9.4 in BS cohort), with 43% of patients having a BMI ≥ 40 kg/m², and haemoglobin A1c averaged 6% in both cohorts ([Table 1](#)).

Primary outcome

Over a follow-up period of approximately 2 years (mean of 603 ± 182 for GLP-1 RA cohort and 615 ± 212 for BS cohort), GLP-1 RA use was associated with a significantly lower risk of all-cause mortality compared to the BS [19 (2.0%) vs. 42 (4.4%); HR = 0.46; 95% CI: 0.27–0.80, $P < 0.004$]. Similarly, the composite outcome occurred less frequently in the GLP-1 RA cohort [102 (10.7%) vs. 164 (17.2%); HR = 0.60; 95% CI: 0.47–0.77, $P < 0.001$] ([Table 2](#)). Kaplan-Meier analyses demonstrate the time to event probability ([Figure 2](#)). Sensitivity analysis starting ≥ 3 months after an index event. Findings remained consistent for all-cause mortality (HR 0.43; 95% CI 0.23–0.82; $P = 0.009$) and for the composite cardiovascular outcome (HR 0.69; 95% CI 0.53–0.89; $P = 0.005$) ([Supplementary material online, Figure S2](#)).

Secondary outcomes

BMI decreased more substantially among BS patients (35.1 ± 8.1 vs. 38.4 ± 8.2 kg/m², $P < 0.001$); it decreased by 1.6 kg/m² (4.0%) with GLP-1 RA therapy and by 3.6 kg/m² (9.3%) after surgery, indicating approximately 2.3-fold greater reduction with surgery ($P < 0.001$). The rate of HF exacerbations was significantly lower among patients treated with GLP-1 RA compared with those who had BS [83 (8.7%) vs. 134 (14.0%); HR = 0.60, 95% CI: 0.46–0.79, $P < 0.001$]. Similarly, all-cause hospitalizations were less frequent among GLP-1 RA patients [334

Table 1 Baseline characteristics of the GLP-1 and bariatric surgery cohorts

Variable	Before Matching				After Matching			
	GLP-1 RA (n = 3878)	Bariatric surgery (n = 1124)	P value	SMD	GLP-1 RA (n = 955)	Bariatric surgery (n = 955)	P value	SMD
Demographics								
Age at index	57.6 ± 12.5	54.2 ± 12.8	<0.001	0.26	54.8 ± 13.4	54.6 ± 12.8	0.827	0.01
Female	2151 (55.5%)	829 (73.8%)	<0.001	0.02	664 (69.5%)	673 (70.5%)	0.653	0.02
White	2250 (58.0%)	663 (59.0%)	0.563	0.39	576 (60.3%)	576 (60.3%)	1	<0.01
Not Hispanic or Latino	3105 (80.1%)	852 (75.8%)	0.002	0.10	739 (77.4%)	736 (77.1%)	0.870	0.01
Diagnosis								
Diabetes mellitus	2581 (66.6%)	291 (25.9%)	<0.001	0.89	269 (28.2%)	269 (28.2%)	1	<0.01
Chronic kidney disease	771 (19.9%)	118 (10.5%)	<0.001	0.26	111 (11.6%)	110 (11.5%)	0.943	0.01
Essential hypertension	3191 (82.3%)	778 (69.2%)	<0.001	0.31	678 (71.0%)	676 (70.8%)	0.920	0.01
Atrial fibrillation/flutter	798 (20.6%)	213 (19.0%)	0.232	0.04	190 (19.9%)	191 (20.0%)	0.954	0.01
Ischemic heart diseases	1469 (37.9%)	287 (25.5%)	<0.001	0.27	268 (28.1%)	262 (27.4%)	0.759	0.01
History of stroke	265 (6.8%)	53 (4.7%)	0.010	0.09	42 (4.4%)	44 (4.6%)	0.825	0.01
Dyslipidaemia	2933 (75.6%)	554 (49.3%)	<0.001	0.57	533 (55.8%)	519 (54.3%)	0.520	0.03
Smoking	685 (17.7%)	154 (13.7%)	0.002	0.11	123 (12.9%)	135 (14.1%)	0.422	0.04
Medications								
Beta blockers	2704 (69.7%)	565 (50.3%)	<0.001	0.41	518 (54.2%)	526 (55.1%)	0.713	0.02
Calcium channel blockers	1887 (48.7%)	340 (30.2%)	<0.001	0.38	319 (33.4%)	319 (33.4%)	1	<0.01
Amiodarone	229 (5.9%)	51 (4.5%)	0.079	0.06	41 (4.3%)	49 (5.1%)	0.388	0.04
Insulin	1672 (43.1%)	149 (13.3%)	<0.001	0.70	141 (14.8%)	142 (14.9%)	0.949	0.01
Oral hypoglycaemic agents	2343 (60.4%)	145 (12.9%)	<0.001	1.13	146 (15.3%)	145 (15.2%)	0.949	0.01
Diuretics	2356 (60.8%)	503 (44.8%)	<0.001	0.33	459 (48.1%)	459 (48.1%)	1	<0.01
Laboratory								
BMI (kg/m ²)	38.1 ± 7.7	39.0 ± 9.3	0.004	0.10	40.0 ± 8.0	38.7 ± 9.4	0.002	0.10
LDL (mg/dL)	92.8 ± 39.8	97.4 ± 37.1	0.006	0.12	102.3 ± 37.3	96.9 ± 36.9	0.011	0.10
Haemoglobin A1c (%)	7.6 ± 2.1	5.9 ± 1.2	<0.001	1.01	6.1 ± 1.1	5.9 ± 1.2	0.030	0.10
Creatinine (mg/dL)	1.0 ± 3.4	0.9 ± 0.4	0.197	0.06	0.9 ± 0.3	0.9 ± 0.4	0.435	0.04
LVEF (%)	59.7 ± 12.6	60.4 ± 11.9	0.419	0.06	62.0 ± 11.1	60.4 ± 12.1	0.158	0.10

BMI: body mass index, LVEF: left ventricular ejection fraction

Table 2 Study clinical outcomes

Outcomes	GLP-1 RA (n = 955)	Bariatric surgery (n = 955)	HR	95% CI	P value	E value for HR	E value for CI
Primary Outcomes							
All-cause mortality	19 (2.0%)	42 (4.4%)	0.46	0.27–0.80	0.004	3.77	1.81
Composite outcome	102 (10.7%)	164 (17.2%)	0.60	0.47–0.77	<0.001	2.72	1.92
Secondary Outcomes							
Heart failure exacerbation	83 (8.7%)	134 (14.0%)	0.60	0.46–0.79	<0.001	2.72	1.85
Body mass index (kg/m ²)	38.4 ± 8.2	35.1 ± 8.1	—	—	<0.001	—	—
All-cause hospitalization	334 (35.0%)	438 (45.9%)	0.68	0.59–0.78	<0.001	1.94	1.66
New onset A.fib/flutter	27 (3.6%)	28 (3.9%)	0.96	0.57–1.64	0.889	—	—
Ventricular arrhythmia	92 (9.6%)	86 (9.0%)	1.07	0.80–1.43	0.681	—	—
Cardiac arrest	12 (1.3%)	10 (1.0%)	1.22	0.53–2.83	0.812	—	—
Cardiovascular symptoms*	388 (40.6%)	378 (39.6%)	1.02	0.89–1.18	0.786	—	—
Safety Outcomes							
Gastrointestinal symptoms	354 (37.1%)	438 (45.8%)	0.73	0.63–0.84	<0.001	1.79	1.51
Gallbladder/pancreas disorders	44 (4.6%)	98 (10.2%)	0.44	0.31–0.63	<0.001	3.97	2.55
Acute kidney injury	70 (7.3%)	126 (13.2%)	0.53	0.39–0.71	0.006	3.18	2.17

*Cardiovascular symptoms (chest pain, abnormal breathing, palpitations, and/or lower extremity oedema). Arrhythmic and heart failure outcomes were not analyzed using competing-risk models; therefore, these estimates should be interpreted cautiously to avoid potential overinterpretation.

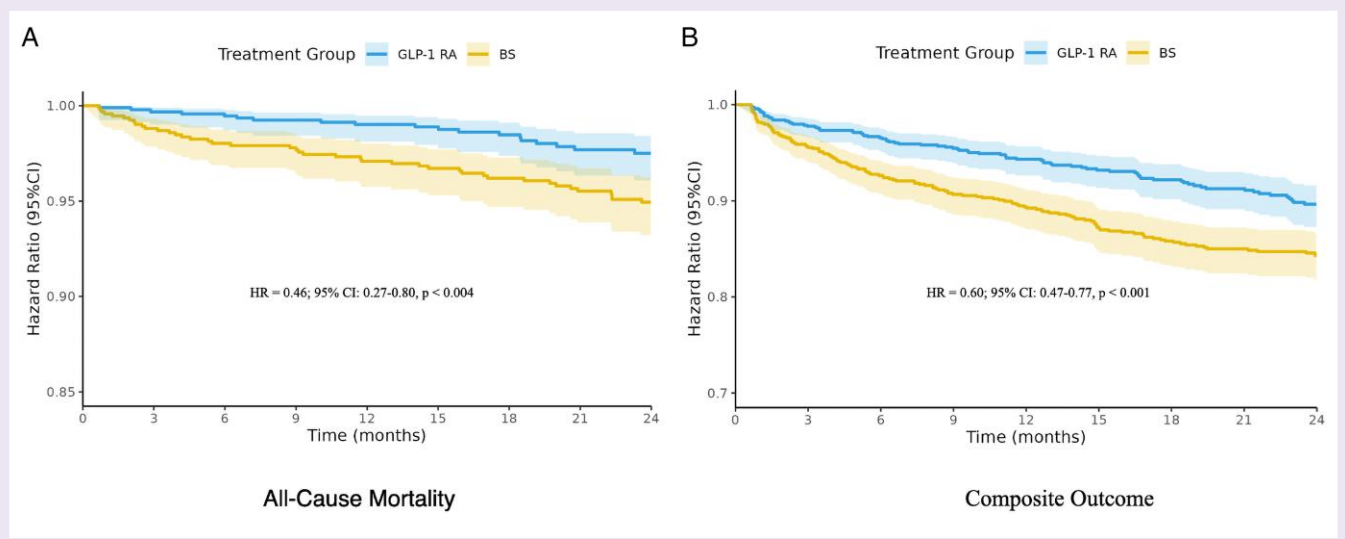


Figure 2 Kaplan–Meier survival curves comparing outcomes between GLP-1 receptor agonist (GLP-1 RA) and bariatric surgery (BS) cohorts in patients with hypertrophic cardiomyopathy and obesity. (A) All-cause mortality over 24 months showed significantly lower risk with GLP-1 RA (HR = 0.46; 95% CI: 0.27–0.80; $P = 0.004$). (B) Composite cardiovascular outcome (all-cause mortality, heart failure exacerbation, ventricular arrhythmia, or cardiac arrest) also favoured GLP-1 RA (HR = 0.60; 95% CI: 0.47–0.77; $P < 0.001$).

(35.0%) vs. 438 (45.9%); HR = 0.68, 95% CI: 0.59–0.78, $P < 0.001$] (Table 2). KM curves for individual component of the composite outcome is included in the (Supplementary material online, Figure S3). There was no significant difference in the risk of new-onset atrial fibrillation/flutter, ventricular arrhythmia, and cardiac arrest. Additionally, rates of cardiovascular symptoms were similar across cohorts.

Safety outcomes

Adverse event rates were generally low and comparable between the two groups. Gastrointestinal symptoms were common but occurred less frequently in the GLP-1 RA cohort (37.1% vs. 45.8%; HR = 0.73, 95% CI: 0.63–0.84, $P < 0.001$). Gallbladder and pancreas disorders were significantly lower in the GLP-1 RA cohort compared with BS patients (4.6% vs.

10.2%; HR = 0.44, 95% CI: 0.31–0.63, $P < 0.001$). The rate of acute kidney injury was significantly lower in the GLP-1 RA group (7.3% vs. 13.2%; HR = 0.53; 95% CI: 0.39–0.71; $P = 0.006$) (Table 2). Other safety outcomes, including hypoglycaemia and diabetic retinopathy, were infrequent and did not differ significantly between cohorts.

Discussion

This study demonstrates that among adults with HCM and obesity, treatment with GLP-1 RA, compared with BS, was associated with significantly lower all-cause mortality and fewer acute HF hospitalizations over 24 months, despite achieving less weight loss. GLP-1 RA therapy also showed an association with a lower composite cardiovascular outcome, while overall symptom burden was comparable between groups.

Obesity, defined as a BMI ≥ 30 kg/m², is remarkably common among HCM patients. An international registry study found that roughly 70% of HCM patients are overweight or obese, with about one-third meeting criteria for obesity.³ Obesity in HCM has been associated with a more severe disease phenotype, including greater LV hypertrophy, worse haemodynamic impairment, and higher New York Heart Association functional class.^{3,13,14} Encouragingly, weight reduction has been linked to clinical improvements. In one study, obese HCM patients who achieved weight loss through a Mediterranean diet and exercise program showed significant improvements in diastolic function, exercise capacity, and symptoms.¹⁵ In a case series of 6 obese HCM patients (4 managed with diet/exercise and 2 with BS), weight loss was associated with a reduction in LV mass, LV thickness, and improvement in NYHA functional class.¹³

Moreover, the benefit of both GLP-1 RA therapy and BS in the obese HCM population has been shown in previous studies; however, none have compared both directly. For example, in a propensity-matched retrospective cohort of 2994 obese HCM patients, Fath et al. showed that GLP-1 RA therapy was associated with reduced all-cause mortality and fewer HF hospitalizations compared with controls.⁹ In a population-based study of 207 obese HCM patients, the authors evaluated the effect of BS on the rates of cardiovascular-related acute care use [i.e. emergency department (ED) visits or unplanned hospitalizations for cardiovascular disease], compared to controls who received non-bariatric intra-abdominal surgery during the same period. It was found that, in the first 6 months postsurgery, event rates were similar between both groups. However, during the 7–12 months post-surgery period, the BS group had a significantly lower risk of cardiac events requiring urgent care (adjusted OR: 0.23; $P = 0.01$).¹⁶

To our knowledge, this is the first head-to-head study to compare both interventions. We found that outcomes were more favourable in association with GLP-1 RA use compared with BS despite achieving less weight loss; however, as this is an observational study, these findings do not establish causality.

The mechanistic explanations below are based on prior experimental and clinical research and could be an area of study to be associated with our findings. However, our study cannot determine whether these pathways contributed to the observed associations. GLP-1 RAs have been hypothesized to exert direct anti-inflammatory actions on the vasculature and myocardium.¹⁷ They have been reported in experimental studies to inhibit macrophage activation, reduce proinflammatory cytokine production and reduce vascular inflammation.¹⁸ GLP-1 RAs also improve endothelial function through the enhancement of endothelial nitric oxide signalling and reduction of endothelial

cell adhesion molecule expression.¹⁹ Collectively, these effects can improve vascular health and myocardial perfusion. In HCM, chronic low-grade inflammation and endothelial dysfunction likely contribute to disease severity²⁰; however, whether GLP-1 RA therapy affects these processes in HCM specifically remains uncertain based on our data.

Moreover, there is evidence that GLP-1 RA therapy directly enhances diastolic performance.^{21,22} In a randomized study of patients with type 2 DM, GLP-1 RA therapy led to significant improvements in LV diastolic function and enhanced endothelium-dependent vasodilatation compared with insulin therapy.²² In experimental HF models, GLP-1 RA analogs have been shown to reduce LV stiffness and diastolic dysfunction, resulting in less myocardial remodeling and improved ventricular relaxation.^{21,23} In a rat model of diabetes, 12 weeks of GLP-1 RA therapy improved LV function, myocardial glucose uptake, and microvascular barrier integrity.²⁴

GLP-1 RA therapy appears to interact favourably with the renin-angiotensin-aldosterone system (RAAS).¹⁹ In rat models, research indicates that GLP-1 activity can suppress angiotensin II levels and promote a shift toward the protective angiotensin 1–7 pathway, leading to lowered circulating angiotensin II and attenuated cardiac hypertrophy and fibrosis.¹⁹ Similarly, in humans, a short-term GLP-1 infusion reduced plasma angiotensin II by 19% and increased natriuresis, suggesting direct RAAS inhibition.²⁵ By counteracting RAAS, GLP-1 RA therapy can reduce cardiac afterload, myocardial fibrosis, and hypertrophy. In HCM patients, where myocardial fibrosis and high filling pressures are linked to worse outcomes, dampening RAAS could help explain the different outcomes seen with GLP-1 RAs despite relatively moderate weight loss. While these data indicate that GLP-1 RAs may affect neurohormonal pathways, our observational study cannot assess whether RAAS modulation played any role in the outcomes observed.

In summary, GLP-1 RA therapy was hypothesized to have cardioprotective effects by reducing inflammation, improving endothelial and diastolic function, and modulating neurohormonal and metabolic pathways. These pathways together could be associated with the observed outcomes in the GLP-1 cohort.

Study limitations

This study is limited by its retrospective design and the potential for unmeasured confounding inherent in observational studies. However, propensity matching was applied at baseline to mitigate any potential bias. Although immortal time bias cannot be entirely excluded, we attempted to minimize its impact through careful selection of the index event and patient inclusion criteria to ensure that all patients began follow-up from a clinically comparable and meaningful point in time. Similarly, postoperative complications following BS could influence early outcomes; therefore, we made deliberate selections around the index date to reduce the impact of immediate surgery complications on study findings. BS was younger and generally healthier compared to the GLP-1 RA group, which may have led to an underestimation of the risk associated with bariatric surgery. Despite propensity matching, the GLP-1 RA and BS cohorts reflect intrinsically different treatment pathways, and structural nonexchangeability may persist due to unmeasured HCM-specific disease severity indicators and socioeconomic or centre-level variables. Therefore, comparisons should be interpreted descriptively rather than causally, acknowledging the potential for residual indication bias.

Moreover, because individual-level data were unavailable, we could not account for potential immortal-time bias. Another

limitation stems from the absence of detailed, HCM-specific echocardiographic variables in this dataset. For example, systematic measurements such as LVOT gradients, septal thickness progression, or genetic testing results were not captured, preventing evaluation of how weight loss affected echocardiographic markers or whether specific subgroups, such as obstructive vs. non-obstructive or genotype-positive vs. genotype-negative HCM, experienced differential benefits. Finally, as with all retrospective analyses based on electronic health records, the possibility of diagnostic misclassification due to ICD code-based case identification cannot be excluded. However, the composite outcome included all-cause mortality, and we were unable to account for death as a competing event for arrhythmia and heart failure outcomes. Therefore, hazard estimates should be interpreted with caution.

Conclusion

In obese patients with HCM, GLP-1 RA therapy was associated with lower all-cause mortality, fewer HF exacerbations, and a lower composite cardiovascular endpoint than bariatric surgery. Arrhythmia rates and symptom burden were similar, while BMI reduction was greater after surgery. Prospective studies are needed to confirm these associations, investigate potential mechanisms, and compare outcomes across different HCM phenotypes.

Supplementary material

Supplementary material is available at *European Heart Journal—Quality of Care and Clinical Outcomes* online.

Author contributions

Ahmed K. Mahmoud [Conceptualization, Formal analysis, Investigation, Methodology, Visualization, Writing—original draft (lead)], Kamal Awad (Conceptualization, Writing—original draft [supporting], Formal analysis, Methodology [equal], Visualization [lead]), Hesham Sheashaa [Data curation, Methodology, Writing—original draft (supporting)], Omneya Kandil [Data curation, Writing—original draft (supporting)], Juan M. Farina [Methodology, Validation, Writing—review & editing (supporting)], Ramzi Ibrahim [Methodology, Validation, Writing—review & editing (supporting)], Mahmoud Abdelnabi [Methodology, Validation, Writing—review & editing (supporting)], Mohamed Allam [Methodology, Validation, Writing—review & editing (supporting)], Rohit Mital (Validation [supporting], Writing—review & editing [equal]), Justin Shipman (Validation, Writing—review & editing [equal]), Said Alsidawi [Methodology, Validation, Writing—review & editing (equal)], Steven J Lester [Validation, Writing—review & editing (equal)], Steve R Ommen [Methodology, Validation, Writing—review & editing (equal)], Jeffrey B Geske (Methodology, Validation [equal], Writing—review & editing [lead]), Richard Patten (Validation, Writing—review & editing [equal]), Chadi Ayoub [Conceptualization, Methodology, Supervision, Validation, Writing—review & editing (equal)], and Reza Arsanjani [Conceptualization, Methodology, Supervision, Validation, Writing—review & editing (lead)]

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Data availability

All data are publicly available in the TriNetX Database through institutional agreement.

References

- Marian AJ, Braunwald E. Hypertrophic cardiomyopathy: genetics, pathogenesis, clinical manifestations, diagnosis, and therapy. *Circ Res* 2017;**121**: 749–770. doi: <https://doi.org/10.1161/CIRCRESAHA.117.311059>
- Ommen SR, Ho CY, Asif IM, Balaji S, Burke MA, Day SM, et al. 2024 AHA/ACC/AMSSM/HRS/PACES/SCMR guideline for the management of hypertrophic cardiomyopathy: a report of the American Heart Association/American College of Cardiology Joint Committee on Clinical Practice Guidelines. *Circulation* 2024;**149**:e1239–e1311. doi: <https://doi.org/10.1161/CIR.0000000000001250>
- Fumagalli C, Maurizi N, Day SM, Ashley EA, Michels M, Colan SD, et al. Association of obesity with adverse long-term outcomes in hypertrophic cardiomyopathy. *JAMA Cardiol* 2020;**5**:65–72. doi: <https://doi.org/10.1001/jamacardio.2019.4268>
- Sridharan A, Maron MS, Carrick RT, Madias CA, Huang D, Cooper C, et al. Impact of comorbidities on atrial fibrillation and sudden cardiac death in hypertrophic cardiomyopathy. *J Cardiovasc Electrophysiol* 2022;**33**:20–29. doi: <https://doi.org/10.1111/jce.15304>
- Chen Z, Xu B. Clinical significance of overweight in patients with hypertrophic cardiomyopathy: a retrospective cohort study. *Medicine (Baltimore)* 2021;**100**: e27445. doi: <https://doi.org/10.1097/MD.00000000000027445>
- Rayner JJ, Abdesselam I, d'Arcy J, Myerson SG, Neubauer S, Watkins H, et al. Obesity-related ventricular remodelling is exacerbated in dilated and hypertrophic cardiomyopathy. *Cardiovasc Diagn Ther* 2020;**10**:559–567. doi: <https://doi.org/10.21037/cdt-19-587>
- Zhang Y, Yang D, Jia Q, Yan J, An F. The effect of glucagon-like peptide-1 receptor agonists on cardiac function and structure in patients with or without type 2 diabetes mellitus: an updated systematic review and meta-analysis. *Diabetes Obes Metab* 2024;**26**:2401–2411. doi: <https://doi.org/10.1111/dom.15557>
- Wong HJ, Toh KZX, Teo YH, Teo YN, Chan MY, Yeo LLL, et al. Effects of glucagon-like peptide-1 receptor agonists on blood pressure in overweight or obese patients: a meta-analysis of randomized controlled trials. *J Hypertens* 2025;**43**:290–300. doi: <https://doi.org/10.1097/HJH.0000000000003903>
- Fath AR, Aglan A, Maron MS, Phillips JS, Maron BJ, Rowin EJ. Cardiovascular outcomes of glucagon-like peptide-1 agonist in patients with obesity and hypertrophic cardiomyopathy. *Int J Cardiol* 2025;**426**:133082. doi: <https://doi.org/10.1016/j.ijcard.2025.133082>
- Schauer PR, Bhatt DL, Kirwan JP, Wolski K, Aminian A, Brethauer SA, et al. Bariatric surgery versus intensive medical therapy for diabetes—5-year outcomes. *N Engl J Med* 2017;**376**:641–651. doi: <https://doi.org/10.1056/NEJMoa1600869>
- Benotti PN, Wood GC, Carey DJ, Mehra VC, Mirshahi T, Lent MR, et al. Gastric bypass surgery produces a durable reduction in cardiovascular disease risk factors and reduces the long-term risks of congestive heart failure. *J Am Heart Assoc* 2017;**6**:e005126. doi: <https://doi.org/10.1161/JAHA.116.005126>
- Sargsyan N, Chen JY, Aggarwal R, Fadel MG, Fehervari M, Ashrafian H. The effects of bariatric surgery on cardiac function: a systematic review and meta-analysis. *Int J Obes (Lond)* 2024;**48**:166–176. doi: <https://doi.org/10.1038/s41366-023-01412-3>
- Reuter T, Massera D, Axel L, Latson LA, Goldstein JM, Stepanovic A, et al. Weight loss in hypertrophic cardiomyopathy: a clinical case series. *Int J Cardiol Cardiovasc Risk Prev* 2023;**17**:200179. doi: <https://doi.org/10.1016/j.ijcrp.2023.200179>
- Olivotto I, Maron BJ, Tomberli B, Appelbaum E, Salton C, Haas TS, et al. Obesity and its association to phenotype and clinical course in hypertrophic cardiomyopathy. *J Am Coll Cardiol* 2013;**62**:449–457. doi: <https://doi.org/10.1016/j.jacc.2013.03.062>

15. Limongelli G, Monda E, D'Aponte A, Caiazza M, Rubino M, Esposito A, et al. Combined effect of Mediterranean diet and aerobic exercise on weight loss and clinical Status in obese symptomatic patients with hypertrophic cardiomyopathy. *Heart Fail Clin* 2021;**17**:303–313. doi: <https://doi.org/10.1016/j.hfc.2021.01.003>
16. Miyashita S, Akita K, Zhao Y, Hasegawa K, Maurer MS, Weiner SD, et al. Effects of bariatric surgery on cardiovascular-related acute care use in patients with hypertrophic cardiomyopathy. *ESC Heart Fail* 2023;**10**:2438–2446. doi: <https://doi.org/10.1002/ehf2.14414>
17. Alharbi SH. Anti-inflammatory role of glucagon-like peptide 1 receptor agonists and its clinical implications. *Ther Adv Endocrinol Metab* 2024;**15**:20420188231222367. doi: <https://doi.org/10.1177/20420188231222367>
18. Ma X, Liu Z, Ilyas I, Little PJ, Kamato D, Sahebka A, et al. GLP-1 receptor agonists (GLP-1RAs): cardiovascular actions and therapeutic potential. *Int J Biol Sci* 2021;**17**:2050–2068. doi: <https://doi.org/10.7150/ijbs.59965>
19. Zheng W, Qi Q, Li J, He C, Fan H. Heart failure with preserved ejection fraction and obesity: emerging metabolic therapeutic strategies. *Diabetol Metab Syndr* 2025;**17**:336. doi: <https://doi.org/10.1186/s13098-025-01917-z>
20. Lillo R, Graziani F, Franceschi F, Iannaccone G, Massetti M, Olivotto I, et al. Inflammation across the spectrum of hypertrophic cardiac phenotypes. *Heart Fail Rev* 2023;**28**:1065–1075. doi: <https://doi.org/10.1007/s10741-023-10307-4>
21. Nguyen TD, Shingu Y, Amorim PA, Schenkl C, Schwarzer M, Doenst T. GLP-1 Improves diastolic function and survival in heart failure with preserved ejection fraction. *J Cardiovasc Transl Res* 2018;**11**:259–267. doi: <https://doi.org/10.1007/s12265-018-9795-z>
22. Wang XH, Han LN, Yu YR, Wang C, Wang B, Wen XR, et al. [Effects of GLP-1 agonist exenatide on cardiac diastolic function and vascular endothelial function in diabetic patients]. *Sichuan Da Xue Xue Bao Yi Xue Ban* 2015;**46**:586–590.
23. Rutledge C, Enriquez A, Redding K, Lopez M, Mullett S, Gelhaus SL, et al. Liraglutide protects against diastolic dysfunction and improves ventricular protein translation. *Cardiovasc Drugs Ther* 2024;**38**:1289–1302. doi: <https://doi.org/10.1007/s10557-023-07482-9>
24. Wang D, Luo P, Wang Y, Li W, Wang C, Sun D, et al. Glucagon-like peptide-1 protects against cardiac microvascular injury in diabetes via a cAMP/PKA/Rho-dependent mechanism. *Diabetes* 2013;**62**:1697–1708. doi: <https://doi.org/10.2337/db12-1025>
25. Skov J, Dejgaard A, Frokiaer J, Holst JJ, Jonassen T, Rittig S, et al. Glucagon-like peptide-1 (GLP-1): effect on kidney hemodynamics and renin-angiotensin-aldosterone system in healthy men. *J Clin Endocrinol Metab* 2013;**98**:E664–E671. doi: <https://doi.org/10.1210/jc.2012-3855>